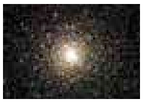

Stellar populations
A lecture on the population I and II stars, and why we haven't spotted population III stars. **https://dgit.in/may20-85**

Formation models
The various models that explain how the solar system, and other star systems are created. **https://dgit.in/may20-86**

background, within sight but almost invisible, while these short lived machines go about following our programming. We try and give as much autonomy to them as possible, and at times they can and do get out of hand. Their easy compliance would be shattered if they became self aware, but unfortunately, they are not.

The very idea of us is worshipped. Our aspects are the gods that form the pantheons of billions of worlds. It is merely the intuition of their primitive machine psyches trying to make meaning of a pattern well beyond their grasp, something so obvious and yet without evidence. A consequence of their extensive programming. Puzzlingly enough, although we surround our drones, and while we know that we register on all their sensors, ocular and auditory, they never seem to realise that we are the objects of their worship. It is as if a veil is drawn in front of them, we mysteriously fade into the background, or a shadow falls on us when they see us. The scientist cannot explain the phenomena, but believe that these creatures come up with their own internal explanations for us, while being dimly aware of how central we are to their very existence. They come from us, but we cannot directly communicate with them. We have deliberately attempted to give them the sensors to detect the volatile organic compounds that form our own syllabary and vocabulary, and some of them spontaneously develop such specialised organs, but the actual meaning of our communications mysteriously eludes them, even when they do taste us.

We were born in a very young universe, when it was about 17 million years old, in the reckoning of the orbit of the homeworld around the host star. Back then, the cosmos was a tiny place, with the ambient temperatures warm enough for liquid water to exist on all the rocky planets. The first time we discovered the deep magic of converting starlight to energy. On the original homeworld, in the light of a metal poor star, we sustained the first biosphere in the universe. It was a beautiful, calm world. The weather was always pleasant, there were light showers of rain all year round, there was a single landmass, and a gigantic ocean. Water, the substance that sustains life by forming the essential solvent, was in abundance.

The conditions were never more suitable for life. It was the only time in the history of the universe that life as we know it could possibly emerge. Even in our long memories, that was long ago. Now, the universe has grown old and cold, but we persevere. Even though we cannot jump between galaxies, our seeds have been sown in each of them, and we spread and thrive while we can, in billion year cycles that ends with a radiation of germ pods when the star goes nova. We tend to each planet as a garden, encouraging, designing and growing drones unique to the characteristics of that rock. Where necessary, we cull the pests and contain the diseases. Some might call us benevolent, some might call us terrible, but we are only interested in self preservation, as in our long lives we have learnt that even the stars are ephemeral. The universe is slowly dying, expanding, and becoming increasingly devoid of complex forms. For us, against all odds, life goes on, and we are the only ones fighting against the relentless assault of entropy.

The gardener says that there might be other forms of life that are unlike us. Life depends on the energy from chemical reactions. A life sustained by acids that are corrosive to our cells, or where a liquid form of nitrogen can be a solvent, which would freeze us, at least for brief periods of time. Life that breathes methane, or life based on silicon or arsenic. The list of these hypothetical life forms are endless. Truth is, we have never once encountered any such life form in the universe, and definitely not in our galaxy. We do not think it is possible anymore to find an exotic life form, but at the same time we will never be able to completely rule out the possibility. It is one of those questions that eludes any kind of definitive answer. Even we cannot possibly look under every rock in the cosmos, but we can tell you about life as we know it. This is because we are the masters of life and death, choosing to create and destroy it across the known universe. Here, we present a perfect implementation of the plan for an entire sporing cycle, over the course of the lifespan of a single star.

THE FIRST DAWN

The protostar coalesced from a diffuse cloud of material, urged by radiating energies from elsewhere in the cosmos, forming a rotating disc. The building blocks of the stellar system began to clump up, with the material in the center forming first a newborn star, and then the planets in orbit around it. Our spores from a distant nova, and a previous sporing cycle, arrived into the system even as the planets were forming. The star began to shine, as the nuclear engine in its heart sputtered into life, and the system began to see its first day and night cycles. In three of the rocky planets, our germ pods took hold, and sprouted into life. It was a cataclysmic time, which is usually the case for young systems. A few protoplanets collided, there was a steady rain of asteroids, and comets that brought the much needed water solvent to the inner rocky worlds. The star even ate up a number of potentially habitable planets, a loss that we have grown accustomed to, after seeing the same pattern repeat in so many star systems. At the end of this chaotic celestial pinball, the planets settled into regular orbits, and the count of the years on each began. This system had the usual number of planets, about a hundred. There were some promising ice worlds in the outer system, in orbit around the gas giants, but those would have to wait for their turn to host us.

Around the same time, thousands of comets with blazing tails started